

The definitive GCP report.

The GCP Event Experiment: Design, Analytical Methods, Results

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Abstract—Studies of anomalous correlations between mind and matter usually focus on participating subjects and isolated target systems. We report on a decade-long experiment which finds that anomalous mind-matter correlations may be a pervasive aspect of reality. The Global Consciousness Project (GCP) measures the output deviation of a global network of physical random number generators (RNGs) at the time of major world events. The project hypothesizes that the coherent attention or emotional response of large populations induced by the events will correspond to characteristic deviations of the network output. We describe the motivation and scope of the experiment and the analytical procedures employed to test the hypothesis, and present the results of 236 events accumulated over the first 9 years of operation. The cumulative significance across all events favors the hypothesis by more than 4.5 standard deviations. Beyond a test of the basic hypothesis, secondary analyses show that the result is driven by correlations in the RNG network across global distances.

Keywords: random numbers—REG—RNG—anomalies—global consciousness—consciousness—parapsychology—mind-matter interaction

1. Introduction

In this paper we present an analysis of an ongoing experiment conducted by the Global Consciousness Project (GCP). The long-term experiment studies the proposition that subtle deviations in random systems will correlate with periods of intense collective behavior in global populations. The project is motivated by extensive experimental evidence documenting anomalous effects in random number generators (RNGs) under conditions of directed mental intention.^{1–3} Such effects have also been described in local field-based studies during group events of a psychologically cohesive nature.^{4–7}

The GCP extends these studies to a global scale by hypothesizing that world events which provoke an emotive or attentional response from large numbers of people will correspond to periods of anomalous deviations in a geographically distributed RNG network. The GCP network consists of over 65 RNGs deployed

at fixed sites around the world. Data from the RNGs are sampled once per second and archived via the Internet into a continuously updated database. Tests of the hypothesis are performed by identifying data periods of pre-specified events and applying pre-specified analysis algorithms. After registration of these analysis parameters, the data archive is opened and a z-score (i.e., the standard normal deviation from expectation) is generated from the predetermined algorithm. Over 250 replications of this protocol have been implemented since the project's inception in 1998, and 236 of these meet strict criteria for network stability and correct hypothesis definition. To date, the cumulative score of data deviations during the designated events stands at 4.5 standard deviations (p-value $\sim 3 \times 10^{-6}$), confirming the general hypothesis to high significance.

The purpose of this paper is to present a full description of these results, including an indication of the structure underlying the statistical deviations we find, and to draw some preliminary conclusions about the experiment's implications for parapsychological or psi research. We also correct some misconceptions and incorrect interpretations of the project that have appeared in the popular press and elsewhere.^{8,9} Finally, we intend this paper as a foundation for a series of detailed investigations to extend and illuminate the primary results.

Parapsychology developed in the nineteenth century to assess the validity of extraordinary anecdotal claims of anomalous perception. Research focused on the examination of case studies, following standard methods practiced in psychology at the time. The extensive literature, although puzzling and controversial, indicated to many researchers that the case studies could not be wholly explained as delusional episodes or misconstrued psychological projections. With the advent of experimental psychology in the twentieth century, research moved to the laboratory where it gradually attracted the interest of scientists from other fields who brought a range of methods and techniques to the problem. One consequence, which is implicit in the locution "anomalous phenomena" as an alternative designation for psi, has been a growing body of evidence that the variety of phenomena investigated cannot be understood only in psychological terms.

Psi phenomena reported in the literature can be framed in two ways: as anomalous perception, by which an individual accesses information inaccessible to the ordinary senses, or as anomalous physical behavior, for which measured deviations from expectation in physical systems remain unexplained by physical laws. Laboratory studies of anomalous perception include telepathy, Ganzfeld and remote-viewing experiments.¹⁰ These often involve a pair of subjects, and a simple conception of positive outcomes is that the anomaly involves an access or sharing of mental contents between the subjects. Similar experiments using a single subject and an external target of some kind suggest that the reach of anomalous perception extends to the environment in a general way. Related studies which monitor physiological responses of subjects indicate that anomalous perception may occur subliminally and need not be accompanied by mentation or conscious reports.^{11,12} Experiments which employ physiological measurements raise the question of how psi phenomena fit with our knowledge

of the material world. Considering these studies, we can ask whether psi should be construed as a mental phenomenon, a mind-matter interaction, or a subtle correlation between the physical and the mental.

A careful parsing of interpretations leads straight into the controversies current in contemporary debates on consciousness and the mind/body problem. This is an inevitable difficulty for research which deals with phenomena anchored in the non-material or mental domain. Indeed, terms such as "mental", "mind", "information" and even "physical" are laden with often implicit or imprecise assumptions. A related difficulty involves how to formulate the relationship between the dual domains indicated by our experiential distinction of the mental and the physical. Different approaches may describe the relationship in terms of causation, interaction, correlation, or epiphenomena. Each of these carries deep assumptions which are often difficult to explicate fully.

Here, we avoid presuppositions about the relation between material and mental domains, and about how our results might be accommodated theoretically. We work from a commonsense view of the distinction between material and mental phenomena, and remain open as to how these concepts may be understood in the light of experimental results. While ontological and epistemological precision is important for the interpretation of data, the adoption of a narrow stance is not required for experimentation, nor is it necessarily desirable at the outset when dealing with such a poorly understood and complex topic as psi. Experiments provide input for models and serve to guide interpretations and shape theory. Accordingly, the GCP experiment aims to test a conjecture which would extend the range of anomalous phenomena currently encompassed by psi research to study an operationally defined "global consciousness". It is, however, premature to regard this as testing a theory of global consciousness.

The experiment is motivated by a large body of laboratory evidence documenting the occurrence of anomalous deviations in physical RNGs.^{2,3} These experiments address the second pole of psi research which investigates how psi manifests in the physical domain. The basic experimental design, developed in the 1960s and refined in the ensuing decades, posits that the statistical output of stable, truly random (typically quantum) systems can be altered by the directed intention of human agents. In a typical experiment, a subject will spend some minutes in the presence of a RNG, often while receiving a sensory feedback of the device output, and mentally "intend" to alter or bias the output in some predetermined way. The experiments find small, but significant deviations which accord with the predetermined human intentions. These results are important because they suggest that anomalous signatures of mental activity may be detectable in the material domain by physical devices. Attractive features of the RNG studies are that they deal with calibrated systems whose physical principles are well understood and that the data analysis employs straightforward statistical methods. A disadvantage is that the effect sizes are small, even by the scale of parapsychological research. A further complication is that the measured effect sizes vary considerably, both within and across studies.

This has led to a lively debate regarding the role that moderator variables such as subject performance, publication bias or RNG characteristics may play in assessing the measured effect.¹³

The issue of subject performance is germane to psi research in general and is closely related to how experiments incorporate and interpret subject intention and the use of "targets". An experimental approach that circumvents the issue altogether proposes that deviations in RNG output may correspond to focused mental activity in groups of people blind to the experiment. Field studies in which a portable RNG is sampled during psychologically engaging group events such as ceremonies, prayer meetings and sporting events have measured significant deviations in RNG variance at the time of the events.⁴⁻⁷ Too few field RNG studies have been reported to assert a local anomalous field RNG effect with complete confidence. Yet, these experiments suggest a link between anomalous phenomena and mental states which extends beyond intentionality or cognition and removes the primacy of the individual subject. In the simplest interpretation, these experiments suggest that collective mental activity is connected in a deep and general way with the physical environment. This contrasts with traditional thinking in parapsychology for which individual cognition, perception and intention are key elements of psi effects. It suggests that anomalous correspondence between the mental and physical domains is not specific to individuals and locales.

Numerous experiments have found positive outcomes while partially relaxing the constraints imposed by intention, subject individuality and target specificity (including the locality this implies). However, a different picture may emerge if all three constraints are removed simultaneously, and this has not been studied in a systematic way. For example, it is less evident how one can attribute a preferred status to either the mental or the physical domain, as is done in some dualist or reductionist frameworks, when both intention and subject individuality are absent. It also challenges a unified understanding of psi by significantly broadening the variety of anomalous phenomena observed.

The GCP extends the RNG design by reducing these constraints as much as possible. The project expands the canonical experiment, in which the focused intention of an individual subject is directed at one RNG, to its most general realization. The individual subject is replaced by large human populations, the single RNG by a synchronous global network, and focused intention is translated into designated periods of collective attention in the population. The experiment then tests for deviations in the network output during the designated times of collective attention. Replicable significant evidence of a correlation between network deviation and collective mental activity would then strengthen the basis for an operational definition of global consciousness.

In the next section we discuss the experimental hypothesis and design, and indicate why we prefer a composite hypothesis to a simple one: the hypothesis is framed broadly and testing is implemented in a series of predictions about specific events. The protocol is described, including the analysis algorithms used

to determine event statistics. In section three we review the technical aspects of the RNG network and data acquisition as well as the data archive structure, normalization and data vetting procedures. The results of the formal event experiment are presented in section four. We show that the highly significant result is due to a broadly distributed effect and is not driven by outliers. An important conclusion is that the effect size per event is small, yielding an average z-score of 0.3 standard deviations. This result, coupled with the clear absence of outliers, means we cannot meaningfully interpret the statistics of individual events, but must rely on composite scores of many events taken together. We argue that the cumulative z-score of the formal experiment is driven by inter-RNG correlations in the network. This is an entirely new result in psi research. The final section discusses these results in light of several interpretational viewpoints and indicates future research directions.

2. Experimental Hypothesis and Design

The GCP hypothesis can be stated as follows:

Periods of collective emotional or attentional behavior in widely distributed populations will correlate with deviations from expectation in a global network of RNGs.

The hypothesis formalizes the guess that a physical system which is a well-defined part of the material world will at times exhibit anomalous behavior correlating with human mental activity. A viable test of the hypothesis requires designation of two elements: 1) an experimental period of mental activity – the event, and 2) a specific measure of deviation for the corresponding data from the RNG network. At the outset, we do not know what the determining factors of the experiment will be. We therefore adopt an approach which avoids over-determining the experimental variables at the start, as is suitable for the beginning stages of an exploratory research endeavor. Emotional or attentional engagement on a global scale is taken as the guiding criterion for event designation. It is obvious that mental activity, both collective and individual, is ubiquitous and ongoing in the world. Nevertheless, a qualitative distinction can be made for events which simultaneously focus the attention of many people separated by regional or global distances. It is reasonable to assume that occasions such as New Year's Eve celebrations or the news report of a major terrorist attack will define global events in this sense, representing identifiably singular instances of synchronous, communal mental activity. The criterion is inclusive in the sense that events with various types of population engagement may be studied in an effort to learn which factors contribute to the hypothesized effect. The experiment is designed to study other potentially influential factors as well. For example, in correspondence with the distributed character of the global events, the RNGs are deployed widely around the globe, which facilitates analyses of the role spatial parameters may play in the effect.

The deviation statistics we employ are measures of the RNG network's

variance. RNG experiments with intending subjects typically measure deviations of the mean, predicting that a bias of the nominally symmetric (about the mean) RNG output will correlate with the subject's intention. This is a sensible protocol because the stated intention distinguishes between outcomes by attributing a preferred "direction" to the experimental system of subject + RNG. When intention (and hence, direction) is not a component of the experimental design, the natural indicator for RNG deviation is a second-order statistic, and canonical variance statistics offer attractive alternatives. It is worth noting that variance is closely related to entropy and that the symmetry of random entropic deviation could be broken if an anomalous coupling to mental phenomena were present. A scenario like this could lend a preferred direction to variance deviations during specified events. The strategy we adopt is to designate a variance measure as the event statistic and to predict a consistent direction of deviation from expectation should an anomalous effect obtain. Specifically, a positive deviation, representing increased variance, is predicted for most of the events. As with the event designation, the criterion for the variance statistic is inclusive. This means that freedom in the choice of the statistic is allowed based on prior experience or an assessment of the event type. The statistic is chosen, of course, prior to examination of the data.

The details of the experimental program can be summarized as follows. Data from a global network of RNGs are acquired continuously into a closed archive. An event of global significance is identified by the project. A time period for the event is determined and a variance statistic defined. The event is designated a *formal event* by entering this information into a hypothesis and prediction registry¹⁴. Appendix 1 shows some examples. After the event is registered the data are unpacked from the archive and the test statistic is calculated. The deviation of the test statistic from expectation is converted to an equivalent normal z-score and the score is added to a table of all formal event results. The experiment seeks to determine whether the composite of all event z-scores differs significantly from the null expectation.

Because the experiment explores new questions, the general hypothesis allows broad latitude in the selection of event variables and as such has limited explanatory power. This is deemed preferable to arbitrarily adopting narrow selection criteria in the absence of experimental or theoretical guidance. It is implicit in the experimental design that a positive cumulative result will provide not only a degree of confirmation for the hypothesis, but will also identify data sets suitable for further analysis. It should be clear that we do not apply inferential hypothesis testing to the individual events; there is no probability acceptance criterion which is applied to the event z-scores.

3. Network and Data Structure

The RNG network has been described previously.^{15,16} Briefly, the network employs research-grade, commercial random bit generators (Orion, Mindsong,



Fig. 1. The global distribution of the GCP network.

PEAR)¹⁷. The devices process quantum-level electronic noise (post-barrier voltage from electron tunneling in diodes or field effect transistors; or Johnson noise) to generate a bit stream with binomial probability of 1/2, at rates of several thousand bits per second.

The RNG circuits are third or fourth generation designs, refined in laboratory research, and quality components are specified to ensure stable long-term operation. They are electromagnetically shielded, and to eliminate residual biases that might nevertheless arise from temperature changes, component aging, or other environmental factors, the bit sequence is processed with a logical exclusive-or operation (XOR) against a known $p = 0.5$ sequence. The Mindsong and PEAR hardware designs include the XOR; it is applied in software to the output of the Orion device. A serial interface is used to connect to the host computer and provide power.

The network consists of approximately 65 RNGs deployed at host computers around the world. The hosts run custom software to acquire data from the local RNG and periodically transfer the checksummed data over the Internet to the project archive in Princeton, NJ. The network hosts are synchronized via an Internet network time protocol and acquire one time-stamped data trial per second from each RNG. The data trials, which are the sum of 200 consecutive XOR'd bits, are collected at the start of each second. Subsequent bits generated during the second are discarded. This is a legacy procedure developed principally at the Princeton Engineering Anomalies Research laboratory¹⁸ which provides additional assurance that consecutive trials are independent and reduces the data to a nominal binomial (200, 1/2) distribution.

The raw data archive is freely available for download at the project website.¹⁹ It contains a continuous record of trials from all commissioned RNGs since the project's initiation on August 5, 1998. As of January 1, 2008 the database

contains over 13 billion trials representing the accumulation of 2.6 terabits of RNG output. The current network deployment is shown in Figure 1.

The data are examined for errors and stability before analysis. Occasional data errors (typically due to electrical supply or serial port malfunction) are easily recognized as sequences of wildly improbable trial values. RNG stability, which is crucial to the experimental design, is verified by a three-pass procedure. First, all trial values with binomial probability $< 10^{-10}$ are considered errors and are removed. This pass finds most of the serious data errors while removing few possibly valid trials (~three out of more than 13 billion). Second, the trials are normalized to zero mean and a variance of one. (The standardization is done separately for each RNG's data since the devices typically have small, characteristic variance biases. The biases are most pronounced for the Mindsong RNGs and appear to be due to a slight negative autocorrelation in the bit output. These biases are on the order of one part in 10,000 and are detectable after some months of continuous data output.) Next, fluctuations in both the mean and variance of the standardized data are examined for each RNG. This is done for fixed time-scales by a selection of data blockings. Blocks with mean or variance exceeding a cutoff value are tagged for more careful examination. The procedure is run for block lengths of 1 minute, 5 minutes, 1 hour, 4 hours, 1 day, 5 days, 1 month and 3 months. A Poisson testing procedure then decides whether tagged blocks should be masked from analysis. If the excluded data for any RNG exceeds 15 minutes on a given day, the data for the day are removed. The third pass re-calculates the normalizations for the vetted data. The blocking procedure is repeated to assure stability for all RNGs at the various time-scales. In practice, we find the devices are stable over years of operation and the infrequent instances of excluded data are usually traced to hardware failure or software problems. The final normalizations produce approximately standard normal trial values (a binomial kurtosis of 2.99 remains²⁰) which can be safely input to analyses.

The deployment of RNGs in the GCP network depends on the availability of host sites as well as local conditions, especially Internet infrastructure. The network has grown over time, but the decommissioning of hosts also causes occasional alterations in the geographical deployment. Local interruptions, which are more frequent in locations where Internet or electrical grid stability is poor, result in intermittent null periods for some RNG sites. Figure 2 shows the evolution of commissioned RNGs over time.

4. The Formal Experimental Results

The experiment currently comprises 236 formal events spanning a period of over 9 years from December, 1998 through January, 2008.²¹ Event durations vary and are typically a day or less with the median length being 4 hours. The event statistics are calculated using one of three different formulations of variance. Two of these comprise 95% of all events and are differentiated by their

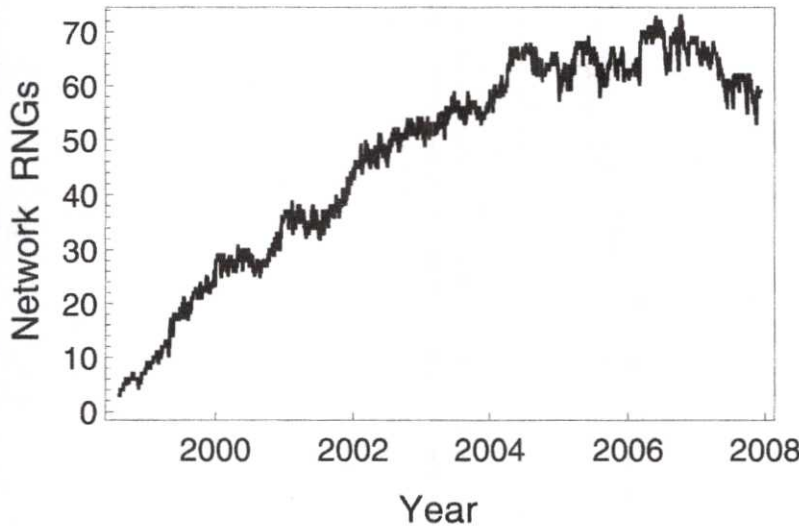


Fig. 2. The evolution of online RNGs in the network with time.

blocking schemes. Most of the remaining events are New Year's Eve celebrations, and use a more elaborate blocking scheme and variance statistic.²²

The two principal variance statistics can be formulated as follows. Normalized trial values are indexed as matrix elements $z_{t,r}$ where r refers to a RNG in the network and t labels the time in seconds. Sums of the trial values follow a normal distribution to high accuracy, by the central limit theorem and the fact that the individual binomial $z_{t,r}$'s closely approximate a normal distribution to begin with. We can convert any such sum to an approximate standard normal variable as:

$$Z_B = \frac{1}{\sqrt{N}} \sum_{t,r}^{T,R} z_{t,r,B} \quad (4.1)$$

where R and T are blockings over RNGs and time, respectively, N is the number of terms in the sum and B is a block index. The variance of Z_B with respect to the (zero) theoretical mean is then

$$\chi^2 = \sum_B Z_B^2 \quad (4.2)$$

which is approximately chi-squared with B degrees of freedom (the standard formulation of variance would divide the chi-squared quantity by B). This formulation provides a compact expression of different variance statistics in terms of the block parameters T and R . The normalization of Z_B by N in Equation 4.1 removes complications from trial vacancies (nulls) when summing over a block. From re-sampling analysis, we find that for $N > 10$, these blocked variance statistics are nearly indistinguishable from a theoretical chi-squared

TABLE 1
The Distribution of Statistical Variance Recipes for 236 Events

Recipe	Time blocking (secs)	Number of events
Network variance (205)	1	186
	60	4
(Epoch)	1	15
Device variance (22)	120	1
	600	1
	900	19
	3600	1
New Year's variance (9)	1	9

Note: The epoch recipes involve a time-aligned concatenation of multiple event periods. Epoch averaging is applied to punctuated events such as New Year's Eve, which is comprised of successive midnights in different time zones. The New Year's variance recipe uses epoch blocking evaluated for a complex recipe devised for New Year's Eve celebrations.

distribution on the time-scale of the events, and the theoretical distribution can be used for assigning probability levels to the event statistics.

Using the blocking formulation, the principal statistics of the formal experiment are the *network variance* (for which each block includes all RNGs) and the *device variance* (for which each block contains one RNG only). These statistics are distinguished by the correlations they can detect. The network variance is sensitive to correlations between RNGs – a spatial characteristic, and the device variance is sensitive to autocorrelations in the RNG outputs – a temporal characteristic. Special cases obtain for the minimal time blocking of 1 second, which eliminates all temporal correlations. The network variance correlations become purely spatial. The device variance at 1-second blocking assigns a single trial to each block and is simply the variance of all trials. This *trial* variance contributes to both formulations regardless of the time resolution. Table 1 lists the distribution of events among the different statistical recipes employed.

The formal experimental result is defined as the *aggregate* of event z-scores. The event z-scores are derived from the signed, one-tailed probability values of the chi-squared variance statistics and are combined as

$$Z_{Tot} = \frac{1}{\sqrt{N_E}} \sum_E Z_E \quad (4.3)$$

which yields a total z-score with equal weighting for events. We find $Z_{Tot} = 4.55$ (p-value 3×10^{-6} , one-tailed). This is the main result of the experiment. It implies that, at significance greater than 4 standard deviations, the experimental hypothesis obtains for the event data. The event effect size (the mean of the event z-scores) is 0.296, with a 95% confidence interval of (0.163, 0.429). Importantly, the result is not driven by outliers. The 10% trimmed mean (removal of 10% of the event z-scores from each end of the z-score distribution) is 0.300 (confidence interval [0.149, 0.452]). The absence of outliers can also be seen by inspecting plots of the z-score distribution. Figures 3 and 4 show two

Fig. 3.

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